

Enqi Zhang

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Education

Nanjing University (NJU): GPA (88/100) 2023.09-2026.06

MS. in Human Geography Nanjing, China

- Key courses: Spatial Data Analysis; Geographic Information Science; Economics Geography

University of Freiburg (Uni Freiburg) 2024.10-2025.03

Exchange Program Freiburg, Germany

Peking University (PKU): GPA (86/100) 2021.09-2023.06

Minor in Economics Beijing, China

- Key courses: Applied Micro-econometrics, Development Economics, Intermediate Micro/Macro Economics

China Agriculture University (CAU): GPA (3.84/4.00) 2019.09-2023.06

BA. in Rural Development Studies Beijing, China

- Key courses: Advanced Math C; Linear Algebra; Probability Theory & Mathematical Statistic; Environment and Development, Participatory Rural Appraisal, International Development Aid

Research Interests

Rural development, food security, nutrition, climate change, spatial economics, Africa.

Skills and Awards

Language: Chinese (Native), English (Fluent, IELTS 7.5), German (A1)

Computer: Office, Python, Stata, ArcGIS

Awards: First-Class Scholarship at China Agricultural University (Top 10%), First-Class Scholarship at Nanjing University (Top 10%), Outstanding Student Award, Anli Fang Scholarship (Top 10%)

Research Experience

Consumers' Participation Preferences in Community-Supported Agriculture (CSA) 2020.12-2021.09

Funded by Beijing College Students' Innovation Program Guiyang, China

- Designed a consumer survey to capture satisfaction with services across different CSA platforms, participation in ecological farming activities, and perceptions of food security.
- Administered an online survey (n=52) in an urban community in Guiyang and conducted offline semi-structured interviews with key stakeholders from six community-based grocery outlets and one company.
- Conducted a comparative case study to examine how “company” versus “rural broker” strategies differ in social ties, coordination practices, and collaborative performance along the CSA supply chain.

The Sensitivity of Rain-fed Agricultural Crop Yields to CO₂ Fertilization in East Africa 2024.12-2025.03

Funded by China National Natural Science Foundation Nanjing, China

- Visualized the results of C-FIX model to stimulate CO₂ fertilization effects using cell data at 0.01° × 0.01° resolution.
- Analyzed the relationship between CO₂ fertilization and crop yields using Pearson correlation, and estimated the sensitivity of crop yields to CO₂ fertilization via multivariate linear regression in Stata.
- Employed the geographical detector model to identify drivers of CO₂ fertilization, focusing on temperature, vapor-pressure deficit, soil moisture, and atmospheric CO₂.

Social Interaction Effect of Small Farmers' Agricultural Production Technical Efficiency 2022.06-2023.08

Funded by China National Natural Science Foundation Beijing, China

- Developed a theoretical framework for social interaction effects grounded in social network theory, drawing

on embeddedness and structural hole perspectives.

- Constructed a household-level network adjacency matrix from *survey-based gift-exchange records* for 203 rural households in a village in Gansu, China.
- Wrote the results of the Moran's I that examined the *spatial autocorrelation* of agricultural technical efficiency, and a *spatial auto-correlation model* that estimated social interaction effects.

Paving the Road to Food Access: Evidence from Oil Price Shocks in Sub-Saharan Africa 2024.09-2026.01
Funded by International Growth Center (online) **Nanjing, China**

- Estimated the impacts of rising transport costs on African local maize prices using *2SLS/IV models in Stata*. Additional analysis examines heterogeneity by land suitability and drought intensity.
- Constructed a network connecting mine sites and ports, and measured the shortest distance from markets to the network using ArcGIS and Python, which is employed as an instrument variable of transport costs.
- Applied cost-allocation model in ArcGIS to map market catchments (land assigned to each market) from 5 raster datasets: roads, rivers/water bodies, land cover, borders, and elevation.

Work Experience

UNICEF Zimbabwe Country Office 2025.04-2026.04
Child Nutrition and Early Childhood Development Fellowship **Harare, Zimbabwe**

- Designed and coordinated the online distribution of "Positive Parenting" surveys (n=112) to employees at Prosperous Lithium Zimbabwe (a mining company) to assess childcare practices and training needs.
- Consolidated an interactive "Positive Parenting" toolkit covering nutrition, social protection, emotional support, and discipline, and coordinated the review process with relevant ministries.
- Supervised national household survey in the Multiple Indicator Cluster Surveys (MICS) and Zimbabwe Livelihoods Assessment (ZimLAC) programmes, focusing on data record, and anthropometry of children under age 5.
- Reviewed a research paper on "The Historical and Contemporary Influence of Droughts on Child Food Poverty and Malnutrition in Zimbabwe", which applied MICS survey data.

Publications and Working Papers

- **Publications** | Zhu, W., Zhang, Z., Zhang, E., & Xu, H. (2025). The sensitivity of rain-fed agricultural crop yields to CO2 fertilization and its driving mechanisms in East Africa. *Environmental Research*, 272, 121164. **(IF=7.7 Accepted)**
- **Publications** | Pang, H., Chen, Q., Zhang, E. (2024). Social interaction effect of small farmers' agricultural production technical efficiency -- Based on an investigation of 203 farmers in Gansu province. *Chinese Journal of Agricultural Resources and Regional Planning*, 45(06), 180-189. **(IF=3.4 Accepted by a Chinese journal)**
- **Working Paper** | Zhang, E., Peng, C. (2025). Paving the Road to Food Access: Evidence from Oil Price Shocks and Maize Market in Sub-Saharan Africa. **(Planned to submit to the Households in Conflict Network (HiCN) and the Social Conflict and Political Economy (SCoPE) Workshop in 2026)**